

Gym Management System

CSE334s (UG2023) - Database Systems

Technical Project Documentation

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Project Theme & Overview

The **Gym Management System** is a desktop application built with **C# Windows Forms** and **Microsoft SQL Server**. It provides a structured interface for gym staff to manage members, trainers, subscriptions, sports, machines, maintenance activities, and related records using secure **ADO.NET** database interactions.

The application supports authentication, record browsing, searching, insertion, updates, deletion, and management of several one-to-many and many-to-many relationships.

Application Interface

The application consists of **10 forms in total**. Six of these are primary management forms accessible from the dashboard, while the others provide authentication and management of specialized sub-records.

Form 1: Login Form

Purpose: Authentication of employees before access to the system is granted.

Functionality

- Accepts **Employee_ID** and password input.
- Checks the **Employee** table using a parameterized SQL query.
- Opens the dashboard and hides the login form when credentials are valid.
- Displays an error message when login fails.

Database Operations: SELECT

Form 2: Main Dashboard

Purpose: Central navigation menu for the main system features.

Functionality

- Displays six navigation buttons: Members, Trainers, Subscriptions, Machines, Sports, and Employees.
- Shows quick dashboard statistics and opens the related form when a button is selected.
- Acts as the main menu after successful authentication.

Database Operations: SELECT

Form 3: Members Form

Purpose: Full management for gym members.

Functionality

- View all member records.
- Search by member ID, first name, last name, or any combination.
- Add a new member with personal and training-related information.
- Update a selected member after loading row data into the form.
- Delete a member and all related phone, practice, and subscription records.
- Open the phone management form for the selected member.

Database Operations: SELECT, INSERT, UPDATE, DELETE

Form 4: Trainers Form

Purpose: Full management for trainers.

Functionality

- View all trainer records.
- Search by trainer ID or first name.
- Add trainers with salary, experience, and start date details.
- Update only the fields that the user has filled in.
- Set related members' `Trainer_ID` values to NULL before deleting a trainer.

Database Operations: SELECT, INSERT, UPDATE, DELETE

Form 5: Subscriptions Form

Purpose: Full management for member subscriptions.

Functionality

- View all subscription records.
- Search by subscription ID, member ID, or employee ID.
- Add subscriptions with linked member and employee references.
- Update & Delete subscription details.

Database Operations: SELECT, INSERT, UPDATE, DELETE

Form 6: Machines Form

Purpose: Full management for gym equipment.

Functionality

- View all machine records.
- Search by machine ID or machine name.
- Add a machine with name, usage, purchase date, and responsible employee.
- Update machine details.
- Delete maintenance records before deleting the machine itself.
- Open the maintenance form for the selected machine.

Database Operations: SELECT, INSERT, UPDATE, DELETE

Form 7: Sports Form

Purpose: Full management for sports and activities offered by the gym.

Functionality

- View all sports.
- Search by sport ID or sport name.
- Add a new sport.
- Update sport name information.
- Delete related practice records before deleting the sport.

Database Operations: SELECT, INSERT, UPDATE, DELETE

Form 8: Employees Form

Purpose: Full management for gym employees (staff).

Functionality

- View all employees with ID, name, job title, and salary.
- Search by employee ID or job title.
- Add a new employee with ID, name, job title, salary, and password; the password defaults to 1234 if left blank.
- Update an employee.

- Delete an employee after setting related **Subscription** and **Machine** references to NULL.

Database Operations: SELECT, INSERT, UPDATE, DELETE

Form 9: Member Phones Form

Purpose: Management of member phone numbers as a multi-valued attribute.

Functionality

- Displays all phone numbers for the selected member.
- Adds a new phone number.
- Deletes the selected phone number.
- Is opened from the Members form using the **Manage Phones** button.

Database Operations: SELECT, INSERT, DELETE

Form 10: Maintenance Form

Purpose: Tracking maintenance history for a selected machine.

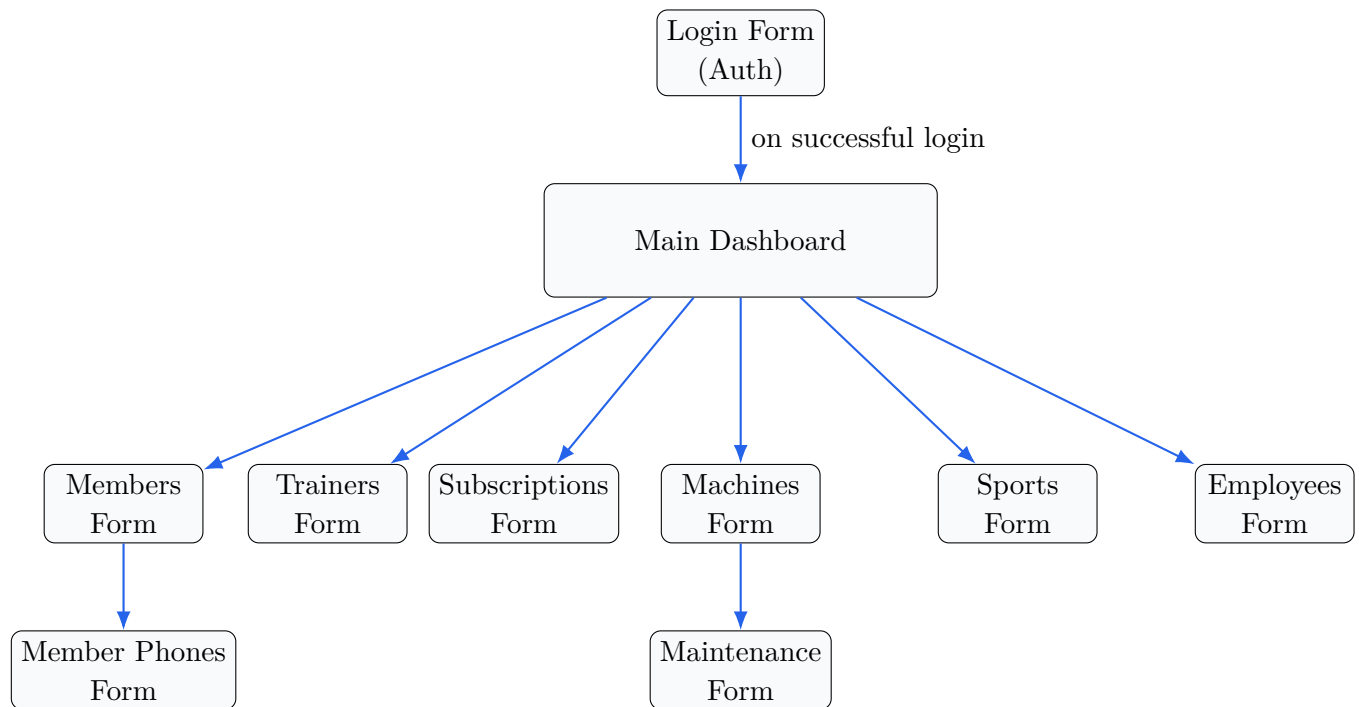
Functionality

- Displays all maintenance records for the selected machine.
- Adds a new maintenance record with start date, end date, and cost.
- Is opened from the Machines form using the **View Maintenance** button.

Database Operations: SELECT, INSERT

Navigation Diagram

The system follows a top-down navigation flow beginning with authentication and continuing to the dashboard and specialized management forms.



Navigation Flow

- `LoginForm` authenticates the employee and opens `MainForm` after a successful login.
- `MainForm` shows summary statistics and contains six buttons that open the Members, Trainers, Subscriptions, Machines, Sports, and Employees forms.
- `MembersForm` opens `MemberPhonesForm` for the selected member.
- `MachinesForm` opens `MaintenanceForm` for the selected machine.